PES-VCS Lab Report

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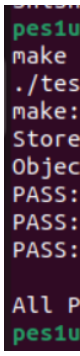
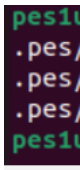
Repository: https://github.com/Shishir-Hegde14/PES1UG24CS438-pes-vcs
Platform: Ubuntu 22.04 (WSL)

Phase 1

Screenshot 1A - `./test_objects` output

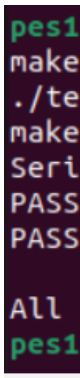


Screenshot 1B - `find .pes/objects -type f`



Phase 2

Screenshot 2A - `./test_tree` output



Screenshot 2B - Raw tree object (`xxd ... | head -20`)

```
pes1ug24cs438@pes1ug24cs438-VirtualBox:~/pes-vcs$ xxd .pes/objects/86/3b36268aa83c8864ed84a1b45785e70fb8fec24a95a36fde2f8f9c61afa5ca | head -20
00000000: 7472 6565 2034 3500 3130 3036 3434 2062  tree 45.100644 b
00000010: 2e74 7874 00e0 0c50 e16a 2df3 8f8d 6bf8  .txt...P.j-...k.
00000020: 09e1 81ad 0248 da6e 6719 f35f 9f7e 65d6  .....H.ng.._.~e.
00000030: f606 199f 7f                             .....
pes1ug24cs438@pes1ug24cs438-VirtualBox:~/pes-vcs$
```

Phase 3

Screenshot 3A - `pes init` -> `pes add` -> `pes status`

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Screenshot 3B - `cat .pes/index`

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pes1ug
```

Phase 4

Screenshot 4A - `pes log` with three commits

```
rm -rf .pes hello.txt bye.txt
./pes init
echo "Hello" > hello.txt
./pes add hello.txt
./pes commit -m "Initial commit"
echo "World" >> hello.txt
./pes add hello.txt
./pes commit -m "Add world"
echo "Goodbye" > bye.txt
./pes add bye.txt
./pes commit -m "Add farewell"
./pes log
Initialized empty PES repository in .pes/
Committed: 6e1e7dd69c68... Initial commit
Committed: 20a42de89b5c... Add world
Committed: 9c1d769dc2f5... Add farewell
commit 9c1d769dc2f57bc5ccea42afed4e729e540f6a213dd9f71cd6270b0f749f73a3
Author: Shishir Hegde <PES1UG24CS438>
Date:   1776430727

    Add farewell

commit 20a42de89b5c9f1c44e2ba8e7e739bb983ec84f29b5c13eaa0d92219953adc8b
Author: Shishir Hegde <PES1UG24CS438>
Date:   1776430727

    Add world

commit 6e1e7dd69c687a267a4fb1c2de19929595715936d339d8aeda491eaedd58306b
Author: Shishir Hegde <PES1UG24CS438>
Date:   1776430727

    Initial commit

pes1ug24cs438@pes1ug24cs438-VirtualBox:~/pes-vcs$
```

Screenshot 4B - `find .pes -type f | sort`

```
pes1
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.pes
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.pes
pes1
```

Screenshot 4C - `cat .pes/refs/heads/main` and `cat .pes/HEAD`

```
pes1ug24cs438@pes1ug24cs438-VirtualBox:~/pes-vcs$ cat .pes/refs/heads/main
cat .pes/HEAD
9c1d769dc2f57bc5ccea42afed4e729e540f6a213dd9f71cd6270b0f749f73a3
ref: refs/heads/main
pes1ug24cs438@pes1ug24cs438-VirtualBox:~/pes-vcs$
pes1ug24cs438@pes1ug24cs438-VirtualBox:~/pes-vcs$
```

Final Integration Test

`make test-integration`

```
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HEAD
ref:
refs
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.pes

===
pes1
```

Phase 5 & 6 Analysis Answers

Q5.1

A branch is a ref file in `.pes/refs/heads/<branch>`, and `HEAD` usually stores `ref: refs/heads/<branch>`.
To implement `pes checkout <branch>`:

1. Verify branch file exists and read its commit hash.
2. Resolve the target commit -> root tree.
3. Compare current working state with index/HEAD to detect conflicts.
4. If safe, rewrite working directory to match target tree:
   - Create needed directories.
   - Write tracked files from blob objects.
   - Remove tracked files/dirs that no longer exist in target tree.
5. Rewrite `.pes/index` to match the checked-out tree metadata.
6. Update `.pes/HEAD` to `ref: refs/heads/<branch>`.

Complexity comes from filesystem synchronization and safety:
- Preserving uncommitted user changes.
- Handling delete/modify conflicts.
- Recursive directory updates.
- Keeping `HEAD`, refs, index, and working tree atomically consistent.

Q5.2

Conflict detection for dirty checkout can be done with index + object store:

1. For each tracked path in index:
   - `stat` current working file.
   - If file missing -> candidate dirty/deleted.
   - If `mtime/size` differ, re-hash file contents and compare with indexed blob hash.
2. If working content differs from index, file is dirty.
3. Compute target branch snapshot membership for each path:
   - If dirty file is also different between current HEAD tree and target tree, it is a checkout conflict.
4. Refuse checkout when any such conflict exists.

This mirrors Git's "would be overwritten by checkout" protection.

Q5.3

Detached HEAD means `HEAD` points directly to a commit hash instead of a branch ref.
New commits are created normally, but no branch pointer moves, so those commits can become unreachable later.

Recovery methods:
1. Immediately create a branch at current commit:
   `git branch rescue-branch <detached-commit>`
2. If already moved away, find commit via reflog-like history (in Git: `git reflog`) and create a branch from that hash.

In PES-style systems, equivalent recovery is creating/updating a ref file to that commit hash before GC removes unreachable objects.

Q6.1

Use mark-and-sweep GC:

1. **Mark roots**: all commits referenced by `.pes/refs/heads/*` (and optionally HEAD if detached).
2. **Traverse reachable graph**:
   - From each commit: mark commit object, then follow `parent` and `tree`.
   - From each tree: mark tree object, then recursively follow subtree/tree entries and blob hashes.
3. Store marked hashes in a hash set for O(1) membership checks.
4. **Sweep** `.pes/objects/*/*`:
   - If object hash not in reachable set, delete it.

For 100,000 commits and 50 branches:
- Commit visits are near the unique commit count, not 50 x 100,000, because branches usually share history.

- Reachable total object visits are approximately: commits + trees + blobs reachable from branch tips.
- In a typical repository this can be several hundred thousand to a few million objects, but traversal remains linear in reachable object count.

Q6.2

Concurrent GC with commit creation is dangerous due races:

1. Commit process writes new blob/tree objects and is about to write commit/ref.
2. GC scans refs before the new commit is referenced.
3. Newly written objects appear unreachable to GC and get deleted.
4. Commit then writes a ref pointing to objects that no longer exist -> repository corruption.

How real Git avoids this:
- Uses lock files/coordination to avoid unsafe overlap.
- Keeps recently created objects protected (mtime grace periods, quarantine, or pack safety rules).
- Performs reference updates atomically.
- Runs GC in ways that do not delete objects that may still be in-flight.